

Week 4 — Probability Review, and the Markowitz Portfolio

AD7031 Optimization Under Uncertainty • Instructor: Prof. H. Park • Sep 19, 2026

1. Events and conditioning

back to week 2's Section 2 table: the OR column had **uncertain data** ξ . today, the vocabulary — written in this course's notation from the first line — x is already taken by the **decision**, so the uncertain data is $\tilde{\xi}$ throughout.

- **random experiment**: its outcome is not known in advance
- **sample space** Ω = the set of all possible outcomes

an **event** is a subset of the sample space, $A \subseteq \Omega$; $\mathbb{P}(A)$ is the probability it occurs.

running example — toss a coin twice: $\Omega = \{HH, HT, TH, TT\}$, each outcome has probability $\frac{1}{4}$

$A = \{\text{first flip is head}\} = \{HH, HT\}$ is an event, $\mathbb{P}(A) = \frac{1}{2}$

- $0 \leq \mathbb{P}(A) \leq 1$, $\mathbb{P}(\Omega) = 1$, $\mathbb{P}(\emptyset) = 0$
- A, B **disjoint** ($A \cap B = \emptyset$): $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$
- **complement** A^c ($A \cap A^c = \emptyset$, $A \cup A^c = \Omega$): $\mathbb{P}(A) + \mathbb{P}(A^c) = \mathbb{P}(\Omega) = 1$
- **monotone**: $A \subseteq B \subseteq \Omega \Rightarrow \mathbb{P}(A) \leq \mathbb{P}(B)$, $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A)$
- A, B **arbitrary**: $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$
— subtract the overlap, or it is counted twice; the disjoint rule is the case $\mathbb{P}(A \cap B) = 0$

Def (*support*) the smallest closed set $\Xi \subseteq \Omega$ such that $\mathbb{P}(\Xi) = 1$ is called the **support** of $\tilde{\xi}$

Def (*conditional probability*) $\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$, defined when $\mathbb{P}(B) > 0$

— B becomes the new sample space; divide by $\mathbb{P}(B)$ so the weights add to 1 again

two coins: $A = \{\text{both flips are heads}\} = \{HH\}$, $B = \{\text{first flip is head}\} = \{HH, HT\}$

$\mathbb{P}(A) = \frac{1}{4}$, $\mathbb{P}(B) = \frac{1}{2}$, $A \cap B = \{HH\}$: $\mathbb{P}(A | B) = \frac{1/4}{1/2} = \frac{1}{2}$

— the first head doubles it, $\frac{1}{4} \rightarrow \frac{1}{2}$

Def (*independent events*) A and B are independent if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$

then $\mathbb{P}(A | B) = \frac{\mathbb{P}(A)\mathbb{P}(B)}{\mathbb{P}(B)} = \mathbb{P}(A)$ — learning B tells you nothing about A

two coins. $A = \{\text{exactly one head}\} = \{HT, TH\}$, $B = \{\text{first flip is head}\} = \{HH, HT\}$

$\mathbb{P}(A \cap B) = \mathbb{P}(\{HT\}) = \frac{1}{4}$, $\mathbb{P}(A)\mathbb{P}(B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$ equal — **independent**

so $\mathbb{P}(A | B) = \frac{1}{2} = \mathbb{P}(A)$: the first flip says nothing about whether there is exactly one head

now $A = \{\text{both heads}\} = \{HH\}$, $B = \{\text{first flip is head}\} = \{HH, HT\}$ — the same test:

$\mathbb{P}(A \cap B) = \mathbb{P}(\{HH\}) = \frac{1}{4}$, $\mathbb{P}(A)\mathbb{P}(B) = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$ $\frac{1}{4} \neq \frac{1}{8}$ — **dependent**

which is why conditioning moved it: $\mathbb{P}(A | B) = \frac{1}{2} \neq \frac{1}{4} = \mathbb{P}(A)$

B did not change, A did — and the verdict flipped. independence is a property of the *pair*, not of either event on its own.

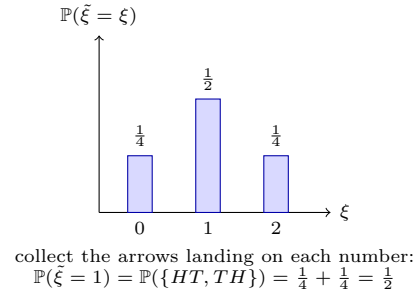
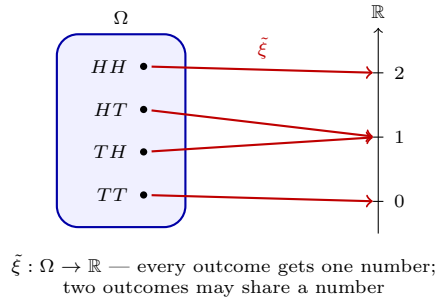
2. Discrete random variables

Def (*random variable*) a function that assigns a real number to each outcome: $\tilde{\xi} : \Omega \rightarrow \mathbb{R}$

the tilde marks the *random* object; ξ without it is a **realization** — one value it took

the same two coins: $\Omega = \{HH, HT, TH, TT\}$, each outcome has probability $\frac{1}{4}$

$\tilde{\xi}$ = number of heads: $\tilde{\xi}(HH) = 2, \quad \tilde{\xi}(HT) = 1, \quad \tilde{\xi}(TH) = 1, \quad \tilde{\xi}(TT) = 0$



- **discrete** $\tilde{\xi}$: countably many values — number of defective parts, number of arrivals, demand
- **continuous** $\tilde{\xi}$: values fill an interval — length, time, temperature, a return

a discrete $\tilde{\xi}$ is a **list**: finitely many values, each carrying a weight.

Def (scenarios) $\tilde{\xi}$ takes the values $\xi^1, \dots, \xi^S$ — the **scenarios** — with

$$p_s = \mathbb{P}(\tilde{\xi} = \xi^s), \quad (1) \quad p_s \geq 0 \quad (2) \quad \sum_{s \in [S]} p_s = 1, \quad [S] = \{1, \dots, S\}$$

$$\mathbb{P}(\tilde{\xi} \in A) = \sum_{s: \xi^s \in A} p_s \quad \text{— add the weights of the scenarios landing in } A$$

the scenario index goes up: ξ^s is scenario s ; a *subscript* ξ_j is component j of a vector (Section 5). two different jobs, two different slots.

older books call this list the **probability mass function** and write it as a function f . same content — but a table, and a table is the only thing a solver can be handed.

$$\text{two coins: } S = 3, \quad \xi^1 = 0, \quad \xi^2 = 1, \quad \xi^3 = 2, \quad p = \left(\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right)$$

$$\mathbb{P}(\tilde{\xi} \geq 1) = p_2 + p_3 = \frac{3}{4} \quad \text{— at least one head, read straight off the list}$$

$$\textbf{Def (mean)} \quad \mu = \mathbb{E}[\tilde{\xi}] = \sum_{s \in [S]} p_s \xi^s$$

$$\text{function of } \tilde{\xi}: \quad \mathbb{E}[h(\tilde{\xi})] = \sum_{s \in [S]} p_s h(\xi^s) \quad \leftarrow \text{law of the unconscious statistician}$$

— keep the weights p_s , change only what they weight. no new distribution needed

Linearity of expectation $\mathbb{E}$ is a weighted sum, so constants pull out and sums split:

$$\mathbb{E}[c] = c, \quad \mathbb{E}[a\tilde{\xi} + b] = a\mathbb{E}[\tilde{\xi}] + b, \quad \mathbb{E}[g(\tilde{\xi}) + h(\tilde{\xi})] = \mathbb{E}[g(\tilde{\xi})] + \mathbb{E}[h(\tilde{\xi})]$$

— each one is $\sum_s p_s(\dots)$ pulled apart; there is nothing deeper in it

$$\textbf{Def (variance)} \quad \sigma^2 = \text{Var}(\tilde{\xi}) = \mathbb{E}[(\tilde{\xi} - \mu)^2] = \sum_{s \in [S]} p_s (\xi^s - \mu)^2, \quad \sigma = \sqrt{\text{Var}(\tilde{\xi})}$$

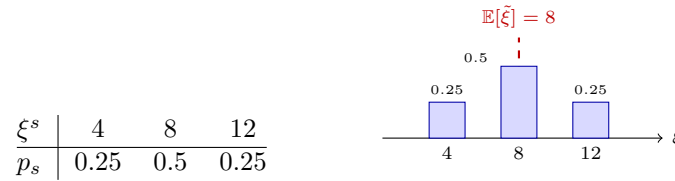
the **shortcut** — expand the square, then apply linearity term by term:

$$\begin{aligned} \text{Var}(\tilde{\xi}) &= \mathbb{E}[(\tilde{\xi} - \mu)^2] = \mathbb{E}[\tilde{\xi}^2 - 2\mu\tilde{\xi} + \mu^2] \\ &= \mathbb{E}[\tilde{\xi}^2] - 2\mu\mathbb{E}[\tilde{\xi}] + \mu^2 = \mathbb{E}[\tilde{\xi}^2] - 2\mu^2 + \mu^2 = \mathbb{E}[\tilde{\xi}^2] - \mu^2 \end{aligned}$$

μ is a number, not random — that is what lets it come out of the $\mathbb{E}$

$$\text{Var}(a\tilde{\xi} + b) = a^2 \text{Var}(\tilde{\xi}) \quad \text{— shifting by } b \text{ moves the centre, not the spread}$$

Example 1 demand $\tilde{\xi}$ for a spare part next month:



$$\mathbb{E}[\tilde{\xi}] = 0.25 \cdot 4 + 0.5 \cdot 8 + 0.25 \cdot 12 = 8$$

$$\text{Var}(\tilde{\xi}) = 0.25 \cdot (4 - 8)^2 + 0.5 \cdot (8 - 8)^2 + 0.25 \cdot (12 - 8)^2 = 8, \quad \sigma = 2\sqrt{2} \approx 2.83$$

3. Continuous random variables

Def (*probability density function, PDF*) $p : \mathbb{R} \rightarrow \mathbb{R}_+$ with

$$(1) \ p(\xi) \geq 0 \quad (2) \ \int_{-\infty}^{\infty} p(\xi) d\xi = 1 \quad (3) \ \mathbb{P}(a \leq \tilde{\xi} \leq b) = \int_a^b p(\xi) d\xi = \text{area under } p \text{ from } a \text{ to } b$$

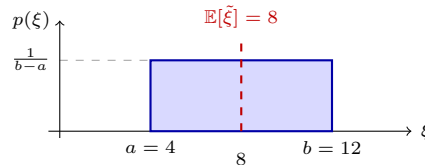
$p(\xi)$ is *not* a probability: $\mathbb{P}(\tilde{\xi} = \xi) = 0$ for every single ξ , and $p(\xi) > 1$ is allowed

Def (*mean, variance*) same as discrete — the sums become integrals:

$$\mu = \mathbb{E}[\tilde{\xi}] = \int \xi p(\xi) d\xi, \quad \sigma^2 = \text{Var}(\tilde{\xi}) = \int (\xi - \mu)^2 p(\xi) d\xi = \mathbb{E}[\tilde{\xi}^2] - \mu^2, \quad \mathbb{E}[h(\tilde{\xi})] = \int h(\xi) p(\xi) d\xi$$

the shortcut $\text{Var}(\tilde{\xi}) = \mathbb{E}[\tilde{\xi}^2] - \mu^2$ carries over unchanged: Section 2 derived it from linearity alone, and linearity holds here for the same reason — the integral splits and constants come out

Example 2 $\tilde{\xi} \sim \text{Uniform}[a, b]$ — $p(\xi) = \frac{1}{b-a}$ on $[a, b]$, 0 outside



mean:

$$\mathbb{E}[\tilde{\xi}] = \int_a^b \xi \frac{1}{b-a} d\xi = \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{a+b}{2}$$

second moment — $h(x) = x^2$:

$$\mathbb{E}[\tilde{\xi}^2] = \int_a^b \xi^2 \frac{1}{b-a} d\xi = \frac{1}{b-a} \left[\frac{x^3}{3} \right]_a^b = \frac{b^3 - a^3}{3(b-a)} = \frac{a^2 + ab + b^2}{3}$$

variance — subtract, no new integral:

$$\text{Var}(\tilde{\xi}) = \mathbb{E}[\tilde{\xi}^2] - \mathbb{E}[\tilde{\xi}]^2 = \frac{a^2 + ab + b^2}{3} - \frac{(a+b)^2}{4} = \frac{(b-a)^2}{12}$$

$$\text{with } a = 4, b = 12: \quad \mathbb{E}[\tilde{\xi}] = 8, \quad \text{Var}(\tilde{\xi}) = \frac{64}{12} = \frac{16}{3}, \quad \sigma = \frac{4}{\sqrt{3}} \approx 2.31$$

Normal distribution $\tilde{\xi} \sim \mathcal{N}(\mu, \sigma^2)$: $p(\xi) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(\xi - \mu)^2}{2\sigma^2}\right), \ \xi \in \mathbb{R}$

$$\mathbb{E}[\tilde{\xi}] = \mu, \quad \text{Var}(\tilde{\xi}) = \sigma^2$$

$$\mathbb{P}(\mu - \sigma \leq \tilde{\xi} \leq \mu + \sigma) \approx 0.68, \quad \mathbb{P}(\mu - 2\sigma \leq \tilde{\xi} \leq \mu + 2\sigma) \approx 0.95, \quad \mu \pm 3\sigma \approx 0.997$$

Six Sigma — yes, the manufacturing one — push the tolerance limits out to $\mu \pm 6\sigma$:

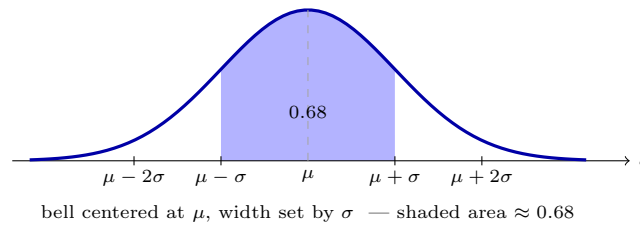
$$\mathbb{P}(\mu - 6\sigma \leq \tilde{\xi} \leq \mu + 6\sigma) = 0.999999998 \quad \text{— about 2 failures per } \textit{billion} \text{ parts}$$

so why is the slogan **3.4 defects per million**, a million times worse? because the standard also allows the process mean to **drift** by up to 1.5σ over a long production run.

the mean slides, the limits do not — the nearer one is then only 4.5σ away, and only that tail matters:

$$\mathbb{P}(\tilde{\xi} \text{ past the nearer limit}) = 3.4 \times 10^{-6} \quad \text{— the famous 3.4 per million}$$

notice the shape of the requirement: *choose the design so that failure has probability at most ε .* that is a **chance constraint**, and it is one of the paradigms this course builds — the same $\tilde{\xi}$, collapsed to a number a different way.



4. Two random variables

one random variable at a time is not enough for Markowitz: n assets, and what makes a portfolio safe is how they move *together*, not how each moves alone.

Def (*independent random variables*) $\tilde{\xi}_1, \tilde{\xi}_2$ are **independent** when their joint law **factors**:

discrete: $\mathbb{P}(\tilde{\xi}_1 = \xi_1, \tilde{\xi}_2 = \xi_2) = \mathbb{P}(\tilde{\xi}_1 = \xi_1) \mathbb{P}(\tilde{\xi}_2 = \xi_2)$

continuous: $p(\xi_1, \xi_2) = p_1(\xi_1) p_2(\xi_2)$

— the same test as independent *events* in Section 1, now asked of every pair of values at once

Def (*covariance*) $\text{Cov}(\tilde{\xi}_1, \tilde{\xi}_2) = \mathbb{E}[(\tilde{\xi}_1 - \mathbb{E}[\tilde{\xi}_1])(\tilde{\xi}_2 - \mathbb{E}[\tilde{\xi}_2])]$

— both above their means together, or both below: positive. one up while the other is down: negative

$\text{Cov}(\tilde{\xi}, \tilde{\xi}) = \text{Var}(\tilde{\xi})$ — variance is the diagonal case

shortcut, the same trick as Section 2: $\text{Cov}(\tilde{\xi}_1, \tilde{\xi}_2) = \mathbb{E}[\tilde{\xi}_1 \tilde{\xi}_2] - \mathbb{E}[\tilde{\xi}_1] \mathbb{E}[\tilde{\xi}_2]$

Def (*correlation*) $\rho(\tilde{\xi}_1, \tilde{\xi}_2) = \frac{\text{Cov}(\tilde{\xi}_1, \tilde{\xi}_2)}{\sigma(\tilde{\xi}_1) \sigma(\tilde{\xi}_2)} \in [-1, 1]$

— covariance carries the units of *both* variables; dividing by the two σ 's strips them out, so ρ compares across pairs

• independent $\Rightarrow \text{Cov}(\tilde{\xi}_1, \tilde{\xi}_2) = 0$: **uncorrelated**

• **the converse is false.** let $\tilde{\xi}$ take $-1, 0, 1$ with probability $\frac{1}{3}$ each, and set $\tilde{\eta} = \tilde{\xi}^2$:

$$\mathbb{E}[\tilde{\xi}] = 0, \text{ so } \text{Cov}(\tilde{\xi}, \tilde{\eta}) = \mathbb{E}[\tilde{\xi} \tilde{\eta}] - \mathbb{E}[\tilde{\xi}] \mathbb{E}[\tilde{\eta}] = \mathbb{E}[\tilde{\xi}^3] = \frac{1}{3}(-1) + \frac{1}{3}(0) + \frac{1}{3}(1) = 0$$

uncorrelated — yet $\tilde{\eta}$ is a *function* of $\tilde{\xi}$: knowing one pins the other exactly.

— covariance sees only the **linear** part of a relationship. zero covariance is not independence

Covariance matrix for a random vector $\tilde{\xi} = (\tilde{\xi}_1, \dots, \tilde{\xi}_n)$: $\Sigma \in \mathbb{R}^{n \times n}$, $\Sigma_{ij} = \text{Cov}(\tilde{\xi}_i, \tilde{\xi}_j)$

symmetric; diagonal $\Sigma_{jj} = \text{Var}(\tilde{\xi}_j)$; and for any weights $x \in \mathbb{R}^n$,

$$\text{Var}(x^\top \tilde{\xi}) = x^\top \Sigma x \geq 0$$

— so $\Sigma \succeq 0$ always. that identity *is* the Markowitz objective; the next section uses it at once

takeaway a portfolio's risk is not the sum of its assets' risks — the off-diagonal Σ_{ij} are exactly what diversification moves.

5. Running example: the Markowitz portfolio

a budget of 1, n assets. decide the split today; the returns are revealed tomorrow.

- **decision:** $x \in \mathbb{R}^n$, x_j = share of the budget in asset j
- **uncertain data:** $\tilde{\xi} = (\tilde{\xi}_1, \dots, \tilde{\xi}_n) \in \mathbb{R}^n$, $\tilde{\xi}_j$ = return of asset j — a random vector with mean μ , covariance matrix Σ
- **feasible set:** $X = \{x \in \mathbb{R}^n : \mathbf{1}^\top x = 1, x \geq 0\}$ ($\mathbf{1}$ = all-ones vector, $\mathbf{1}^\top x = x_1 + \dots + x_n$)

Random vectors the data is a vector now, so the mean is one too and the spread is a matrix:

mean vector $\mu \in \mathbb{R}^n$, $\mu_j = \mathbb{E}[\tilde{\xi}_j]$

$\mu \in \mathbb{R}^n$ and $\Sigma \in \mathbb{R}^{n \times n}$ are the mean vector and the covariance matrix of Section 4 — $\Sigma_{ij} = \text{Cov}(\tilde{\xi}_i, \tilde{\xi}_j)$, symmetric, $\Sigma \succeq 0$

the one fact the model runs on — for a fixed $c \in \mathbb{R}^n$:

$$\mathbb{E}[c^\top \tilde{\xi}] = c^\top \mu, \quad \text{Var}(c^\top \tilde{\xi}) = c^\top \Sigma c$$

— the first is linearity (Section 2); the second is linearity used twice, on a product of two terms

Portfolio return $\tilde{\xi}^\top x = x_1 \tilde{\xi}_1 + \dots + x_n \tilde{\xi}_n$ — a random variable

as a function: fix x , $f(\xi) = \xi^\top x$, $f : \mathbb{R}^n \rightarrow \mathbb{R}$ — the week 2 note's inner product, fed a random input

loss: $f(x, \tilde{\xi}) = -\tilde{\xi}^\top x$ (a gain is a negative loss)

the fact above with $c = x$: $\mathbb{E}[\tilde{\xi}^\top x] = \mu^\top x$, $\text{Var}(\tilde{\xi}^\top x) = x^\top \Sigma x$

Check — two assets the joint distribution below, half the budget in each: $x = (\frac{1}{2}, \frac{1}{2})$

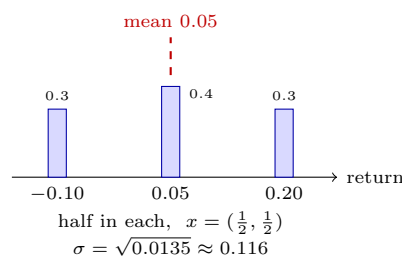
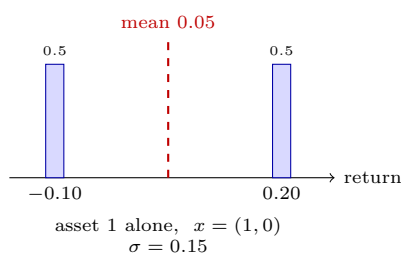
ξ_1	ξ_2	$\frac{1}{2}\xi_1 + \frac{1}{2}\xi_2$	probability	$\Rightarrow$ PMF of $\tilde{\xi}^\top x$:	value	probability
-0.10	-0.10	-0.10	0.3		-0.10	0.3
-0.10	0.20	0.05	0.2		0.05	0.4
0.20	-0.10	0.05	0.2		0.20	0.3
0.20	0.20	0.20	0.3			

from the PMF: $\mathbb{E}[\tilde{\xi}^\top x] = 0.3(-0.10) + 0.4(0.05) + 0.3(0.20) = 0.05$

$$\text{Var}(\tilde{\xi}^\top x) = 0.3(-0.15)^2 + 0.4(0)^2 + 0.3(0.15)^2 = 0.0135$$

from μ and Σ : $\mu^\top x = \frac{1}{2}(0.05) + \frac{1}{2}(0.05) = 0.05 \checkmark$

$$x^\top \Sigma x = \frac{1}{4}(0.0225) + \frac{1}{4}(0.0225) + 2 \cdot \frac{1}{4}(0.0045) = 0.0135 \checkmark$$



same mean, less spread: **diversification**. it comes from $\rho < 1$ — with $\rho = 1$, $x^\top \Sigma x$ would stay 0.0225.

The model (Markowitz, 1952) — least risk, subject to an expected-return target r :

$$\begin{aligned} \min_x \quad & x^\top \Sigma x && \text{variance of the return (risk)} \\ \text{s.t.} \quad & \mu^\top x \geq r && \text{expected return meets the target} \\ & \mathbf{1}^\top x = 1, \quad x \geq 0 \quad x \in X \end{aligned}$$

the week 2 note's QP (its Section 5) with $Q = 2\Sigma$, $c = 0$; convex because $\Sigma \succeq 0$ (its Section 6).

the loss $-\tilde{\xi}^\top x$ is random; Markowitz collapses it with two numbers —
its variance (objective) and its mean (constraint).

Data — the coding session $n = 3$, $r = 0.10$

asset j	expected return μ_j	variance Σ_{jj}	$\sigma_j = \sqrt{\Sigma_{jj}}$
bond	0.06	0.010	0.10
tech	0.10	0.040	0.20
energy	0.15	0.090	0.30

$$x = \begin{pmatrix} x_{\text{bond}} \\ x_{\text{tech}} \\ x_{\text{energy}} \end{pmatrix}, \quad \mu = \begin{pmatrix} 0.06 \\ 0.10 \\ 0.15 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 0.010 & 0.001 & 0 \\ 0.001 & 0.040 & 0.015 \\ 0 & 0.015 & 0.090 \end{pmatrix}$$

covariances: bond–tech 0.001, tech–energy 0.015, bond–energy 0

written out term by term — exactly the lines typed in the code:

$$\begin{aligned} \min \quad & 0.010 x_{\text{bond}}^2 + 0.040 x_{\text{tech}}^2 + 0.090 x_{\text{energy}}^2 + 2(0.001) x_{\text{bond}} x_{\text{tech}} + 2(0.015) x_{\text{tech}} x_{\text{energy}} \\ \text{s.t.} \quad & 0.06 x_{\text{bond}} + 0.10 x_{\text{tech}} + 0.15 x_{\text{energy}} \geq 0.10 \\ & x_{\text{bond}} + x_{\text{tech}} + x_{\text{energy}} \leq 1 \\ & x_{\text{bond}}, x_{\text{tech}}, x_{\text{energy}} \geq 0 \end{aligned}$$

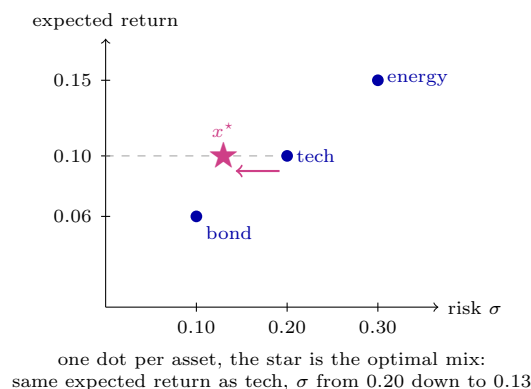
budget “ ≤ 1 ”: leftover sits in cash earning 0 — never helps, so the sum is 1 at the optimum

Solution

$$\boxed{x^* \approx (0.39, 0.29, 0.31), \quad x^{*\top} \Sigma x^* \approx 0.0169, \quad \sigma \approx 0.130} \quad \mu^\top x^* = 0.10 \text{ — target met exactly}$$

all in tech: same expected return 0.10, but $\sigma = 0.20$. the mix cuts σ by about a third.

every asset gets a share — not a vertex (week 2 note, Section 5).



demo `week03_markowitz.ipynb` — Colab — build this model live from the skeleton, then compare with the completed cell.

demo `demos/markowitz_portfolio.py` — the same model as a plain script: prints the model, solves it, plots the weights x^* .

takeaway the running example: the loss $-\tilde{\xi}^\top x$ stays; each later model changes only how it is collapsed to a number — today variance, next CVaR and VaR.

next no class Sep 26. week 6 — duality: every LP comes with a shadow LP, and why $\max = \min$.